

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Mathematics

2. Subject Name: Real Analysis

L-T-P: 3-1-0

Credits: 4

3. Pre-requisites: (i) Single variable calculus, (ii) Multivariable calculus

4. Syllabus and reference books:

Syllabus:

1. Metric Space

- a. Definition, examples, open and closed sets
- b. Sequences, Cauchy sequences and completeness
- c. Continuous and uniformly continuous functions
- d. Compactness and connectedness
- e. $\mathbb{R}^n$ as a metric space: Heine-Borel Theorem, Bolzano-Weierstrass Theorem
- f. Baire's theorem, Cantor Set

2. Measure Theory

- (a) Algebra, σ -algebra
- (b) Outer measure, measurable sets and Lebesgue measure,
- (c) Existence of a non measurable set.
- (d) Measurable functions
- (e) Littlewood's three principles

3. Lebesgue Integration

- a. The Lebesgue integral of a bounded measurable function
- b. The integral of a nonnegative measurable function
- c. The general Lebesgue integral
- d. Convergence theorems: Fatou's Lemma, Monotone convergence and dominated convergence theorem
- e. L^p spaces, Fubini's Theorem

- **Tutorial:** Around 12 tutorial classes will be taken.

Reference Books:

1. Rudin, W. (1964). *Principles of mathematical analysis* (Vol. 3). New York: McGraw-hill.
2. Goldberg, R. R. (1970). *Methods of real analysis*. Oxford and IBH Publishing.
3. Royden, H. L., and Fitzpatrick, P. M. (2010). *Real analysis*. China Machine Press.
4. Stein, E. M., and Shakarchi, R. (2009). *Real analysis: measure theory, integration, and Hilbert spaces*. Princeton University Press.
5. Tao, T. (2011). *An introduction to measure theory* (Vol. 126). American Mathematical Soc..

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Metric Space	14
2.	Measure Theory	13
3.	Lebesgue Integration	13
Total number of hours		40